



STEM SMART Single Maths 12 - Introducing Integration >

Introducing Integration

A-level Maths Topic Summaries - Calculus

Subject & topics: Maths | Calculus | Integration Stage & difficulty: A Level P2

Fill in the blanks below to complete these summary notes introducing integration.

Part A

Indefinite integrals

We use integration to reverse the process of differentiation. The notation $\int f(x) \, dx$ means "integrate the function $f(x)$ with respect to x ".

Complete the following table of indefinite integrals, where a is a constant:

Function, $f(x)$	Integral, $\int f(x) \, dx$
a	
$ax^n, n \neq -1$	
0	

Indefinite integrals always include a , usually written as c (as in the table above) or k .

Items:

-
-
-
-

Part B

Definite integrals

Definite integrals have the form $\int_a^b f(x) \, dx$, where a is the and b is the .

Evaluating definite integrals is a two-step process. We integrate the function, and then we apply the limits.

Example: Evaluate $\int_1^3 5x^2 \, dx$.

First, we integrate the function. This step is the same as for indefinite integrals. We write the result in square brackets. By convention we leave out the constant of integration as this will always cancel out in the next step.

$$\int_1^3 5x^2 \, dx = \left[\right]_1^3$$

Second, we apply the limits. We evaluate the expression in square brackets for the upper limit, then subtract the expression in square brackets evaluated for the lower limit.

$$\begin{aligned} \int_1^3 5x^2 \, dx &= \frac{5}{3} \left[\right]^3 - \frac{1}{3} \left[\right]^3 \\ &= 45 - \frac{1}{3} \\ &= \frac{134}{3} \end{aligned}$$

Items:

1

3

$\frac{5}{3}x^3$

lower limit

upper limit

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Integration (powers of x) 4ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

Part A

Cubic

Find $\int (x^3 + 8x - 5) \, dx$.

The following symbols may be useful: C, c, k, x

Part B

Square root

Find $\int 12\sqrt{x} \, dx$.

The following symbols may be useful: C, c, k, x

Used with permission from UCLES, A Level Maths, January 2009, OCR C2, Question 1

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Indefinite Integrals 1

Pre-Uni Maths for Sciences K1.1

Subject & topics: Maths | Calculus | Integration

Stage & difficulty: A Level P1

Find the following indefinite integrals.

Part A

Integrate $(3x - 1)(x + 1)$

Find $\int (3x - 1)(x + 1) \, dx$.

The following symbols may be useful: C, c, k, x

Part B

Integrate $(\sqrt{p} - \frac{1}{p})^2$

Find $\int \left(\sqrt{p} - \frac{1}{p} \right)^2 \, dp$.

The following symbols may be useful: C, c, k, p

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Indefinite Integrals 2

Pre-Uni Maths for Sciences K1.2

Subject & topics: Maths | Calculus | Integration

Stage & difficulty: A Level P1

Find the following indefinite integrals.

Part A

Integrate $\frac{q^2+3}{q^{\frac{5}{2}}}$

Find $\int \frac{q^2+3}{q^{\frac{5}{2}}} dq$.

The following symbols may be useful: C, c, k, q

Part B

Integrate $2z(z^2-1)(z^2+1)$

Find $\int 2z(z^2-1)(z^2+1) dz$.

The following symbols may be useful: C, c, k, z

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Integrating Powers 1

Pre-Uni Maths for Sciences K1.8

Subject & topics: Maths | Calculus | Integration

Stage & difficulty: A Level P1

Part A

Integrate $4x^3$

Find the indefinite integral of $4x^3$.

The following symbols may be useful: C , c , k , x

Part B

Integrate αx

Find $\int_0^{x_0} \alpha x \, dx$, where α is a constant.

The following symbols may be useful: α , x_0

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Integrating Powers 2

Pre-Uni Maths for Sciences K1.9

Subject & topics: Maths | Calculus | Integration Stage & difficulty: A Level P1

Part A

Integrate ax^{-8}

Find the indefinite integral of ax^{-8} , where a is a constant.

The following symbols may be useful: C, a, c, k, x

Part B

Integrate $\frac{4}{x^2}$

Find $\int_1^2 \frac{4}{x^2} dx$.

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Integration (powers of x) 1ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

Part A

Find integral

Find $\int (2x + 1)(x + 3) \, dx$

The following symbols may be useful: C, c, k, x

Part B

Evaluate integral

Evaluate $\int_0^9 \frac{1}{\sqrt{x}} \, dx$.

Used with permission from UCLES, A Level Maths, June 2005, OCR C2, Question 3

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Definite Integrals 3

Pre-Uni Maths for Sciences K1.5

Subject & topics: Maths | Calculus | Integration **Stage & difficulty:** A Level P1

For the integrals given below, find the values of a and m respectively.

Part A

$$\int_{-1}^a x^{-\frac{4}{5}} dx = 10$$

Given that the definite integral $\int_{-1}^a x^{-\frac{4}{5}} dx = 10$, find the value of a .

The following symbols may be useful: a

Part B

$$\int_0^4 \frac{m+1}{2} x^m dx = 4$$

Given that the integral $\int_0^4 \frac{m+1}{2} x^m dx = 4$, find the value of m .

The following symbols may be useful: m

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Definite Integrals 2

Pre-Uni Maths for Sciences K1.4

Subject & topics: Maths | Calculus | Integration

Stage & difficulty: A Level P1

Find the following definite integrals.

Part A

$$\int_1^{36} \frac{9}{5x\sqrt{x}} \, dx$$

Find $\int_1^{36} \frac{9}{5x\sqrt{x}} \, dx$.

Part B

$$\int_{-8}^0 \frac{1}{\sqrt[3]{x}} \, dx$$

Find $\int_{-8}^0 \frac{1}{\sqrt[3]{x}} \, dx$.

Part C

$$\int_{-1}^1 \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6}\right) \, dx$$

Find $\int_{-1}^1 \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6}\right) \, dx$. Give your answer as an improper fraction.

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Integrating Powers 4

Pre-Uni Maths for Sciences K2.10

Subject & topics: Maths | Calculus | Integration

Stage & difficulty: A Level C2, Further A P2

Part A

Integrate $\frac{A}{r^7} - \frac{B}{r^{13}}$

Find $\int_a^\infty \left(\frac{A}{r^7} - \frac{B}{r^{13}} \right) dr$.

(The force between, for example, two atoms of an inert gas, a distance r apart is given by $\left(\frac{A}{r^7} - \frac{B}{r^{13}} \right)$, where A and B are (negative) constants; the first term is the attractive force between them (the van der Waals interaction, due to their fluctuating induced dipoles) and the second is the repulsive force due to the overlap of their electron shells. The integral describes the potential energy of such a system i.e. the work done bringing one atom from infinity to within a distance a of the other atom.)

The following symbols may be useful: A , B , a

Part B

Integrate $\frac{C}{x^2} + D$

Find $\int_{x_1}^{x_2} \left(\frac{C}{x^2} + D \right) dx$.

(The function $\left(\frac{C}{x^2} + D \right)$, where C and D are constants, could describe the component of an electric field in the x -direction due to a combination of the field due to a point charge at the origin and a uniform field in the x -direction. The integral is then the potential difference between two points x_1 and x_2 on the x -axis.)

The following symbols may be useful: C , D , x_1 , x_2

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